GenCourseAI

Quantum Mechanics for Beginners

An intuitive introduction to subatomic physics, wave-particle duality, superposition, and quantum entanglement.

CURRICULUM COMPILED DYNAMICALLY BY **GENCOURSE AI** ENGINE

Table of Contents

MODULE 1

Module 1: The Quantum Revolution

1.1 Wave-Particle Duality

LESSON OBJECTIVES

- Understand the core concepts of 1.1 Wave-Particle Duality
- Learn the best practices of 1.1 Wave-Particle Duality
- Apply 1.1 Wave-Particle Duality in real-world scenarios

Light and matter exhibit properties of both waves and particles depending on how they are measured. The famous Double-Slit Experiment proved that electrons, traditionally thought of as solid particles, create interference patterns like waves when not observed.

1.2 Superposition & Schrödinger’s Cat

LESSON OBJECTIVES

- Understand the core concepts of 1.2 Superposition & Schrödinger’s Cat
- Learn the best practices of 1.2 Superposition & Schrödinger’s Cat
- Apply 1.2 Superposition & Schrödinger’s Cat in real-world scenarios

Quantum systems exist in a combination of multiple states simultaneously until a measurement occurs. Schrödinger’s Cat is a thought experiment showing the paradox of a cat being both alive and dead in a sealed box until observed.